



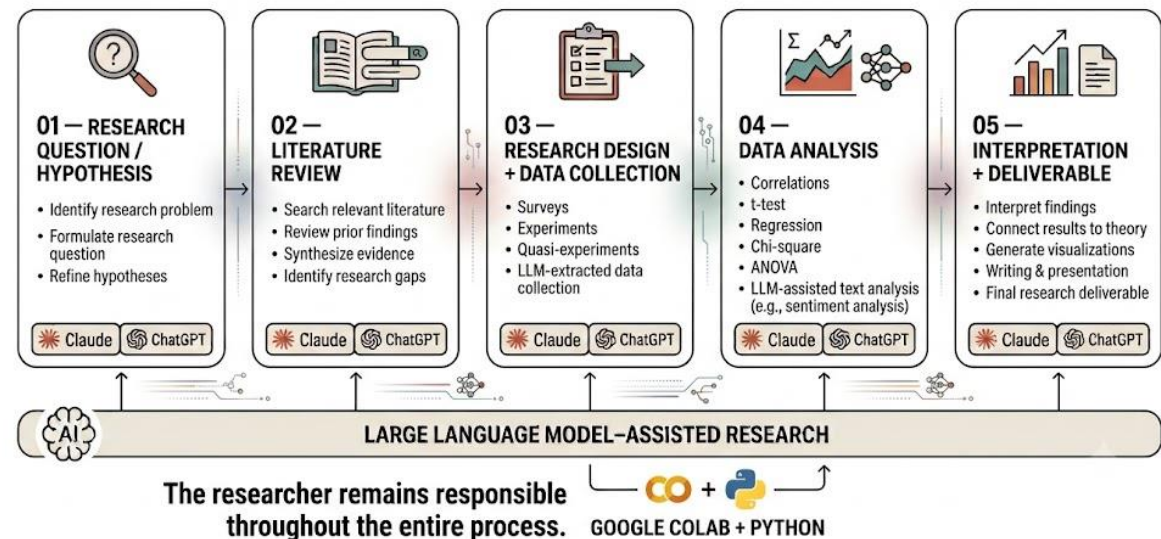
LLM Foundations II: How Does ChatGPT Actually Work?

A simple look inside Large Language Models

Why Look Inside ChatGPT?

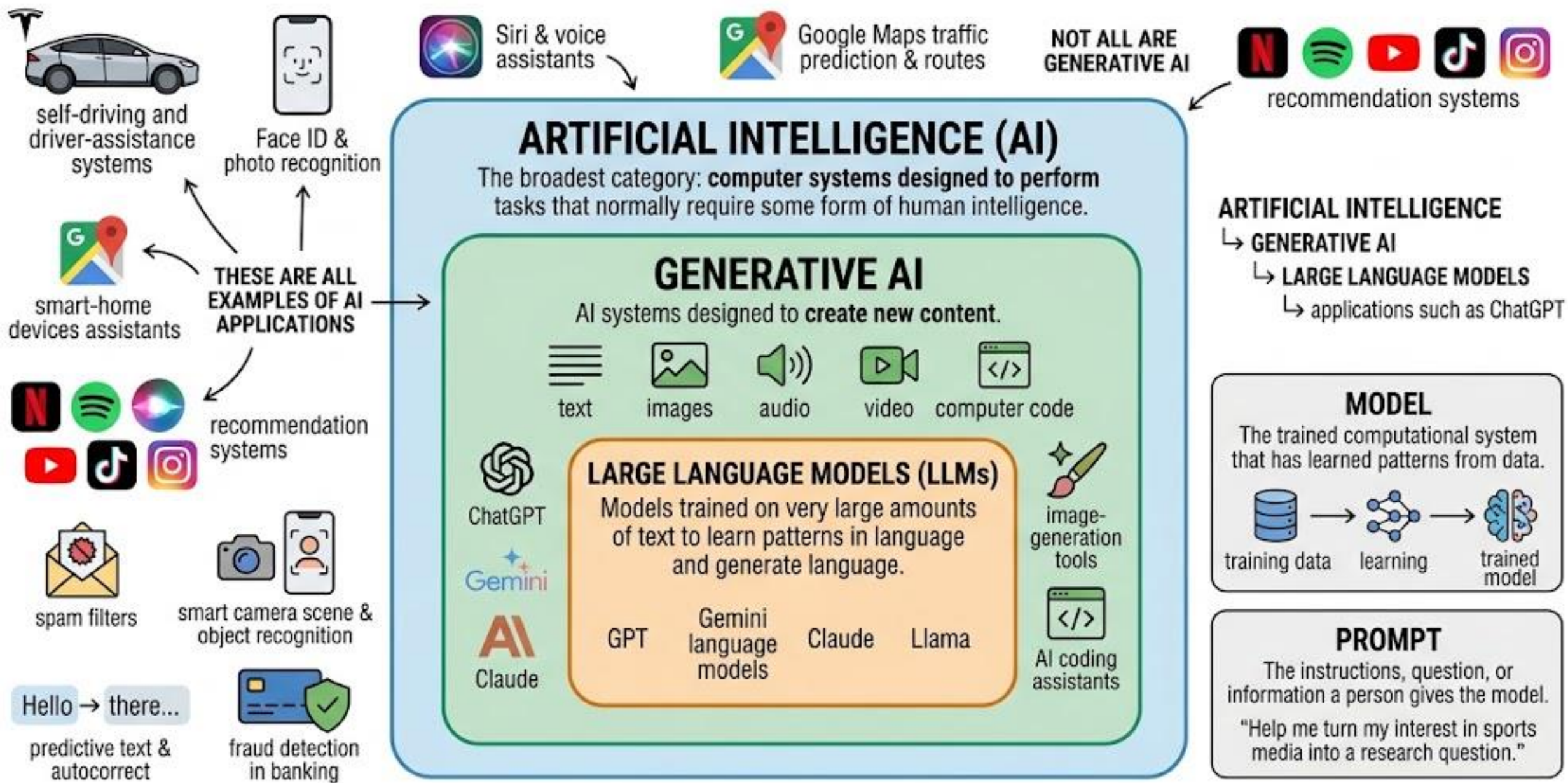
- ChatGPT is a tool we may use in research
- Knowing a little about how it works helps us understand **why it gives the answers it gives**
- It also helps us understand **why it can make mistakes**
- This helps us use AI more carefully in research
- **Researchers should evaluate AI outputs, not just accept them**

Research Methods in the Age of AI



Before We Start: Remember These Words?

- Artificial Intelligence
- Generative AI
- NLP (Natural Language Processing)
- LLM (Large Language Model)
- Token



AI ≠ ChatGPT

ChatGPT is one application that allows people to interact with AI models. AI also includes **recommendation systems**, **computer vision**, **autonomous vehicles**, **prediction systems**, **voice assistants**, and many other technologies.

Remember: How Does an LLM Work?



TRAINING TIME

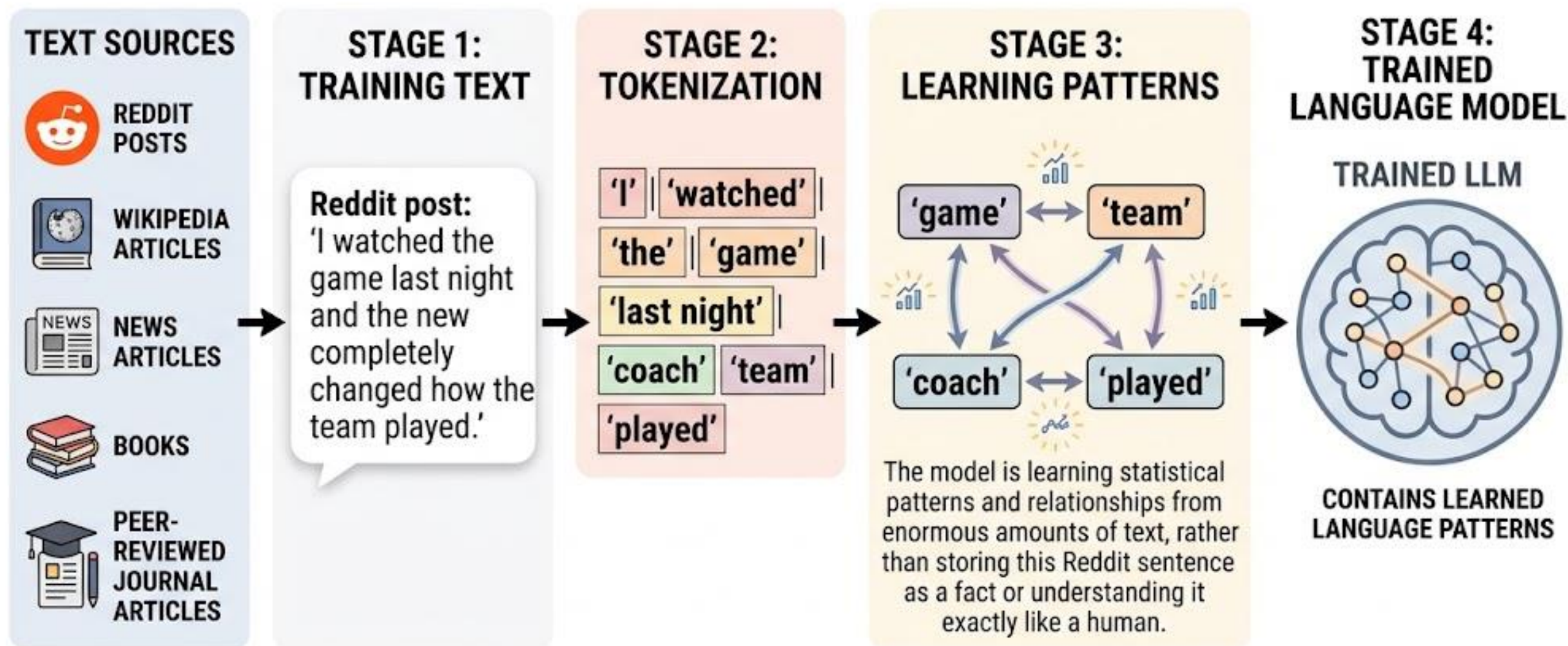


USER TIME

1. Training Time: The Model Is Learning From Text

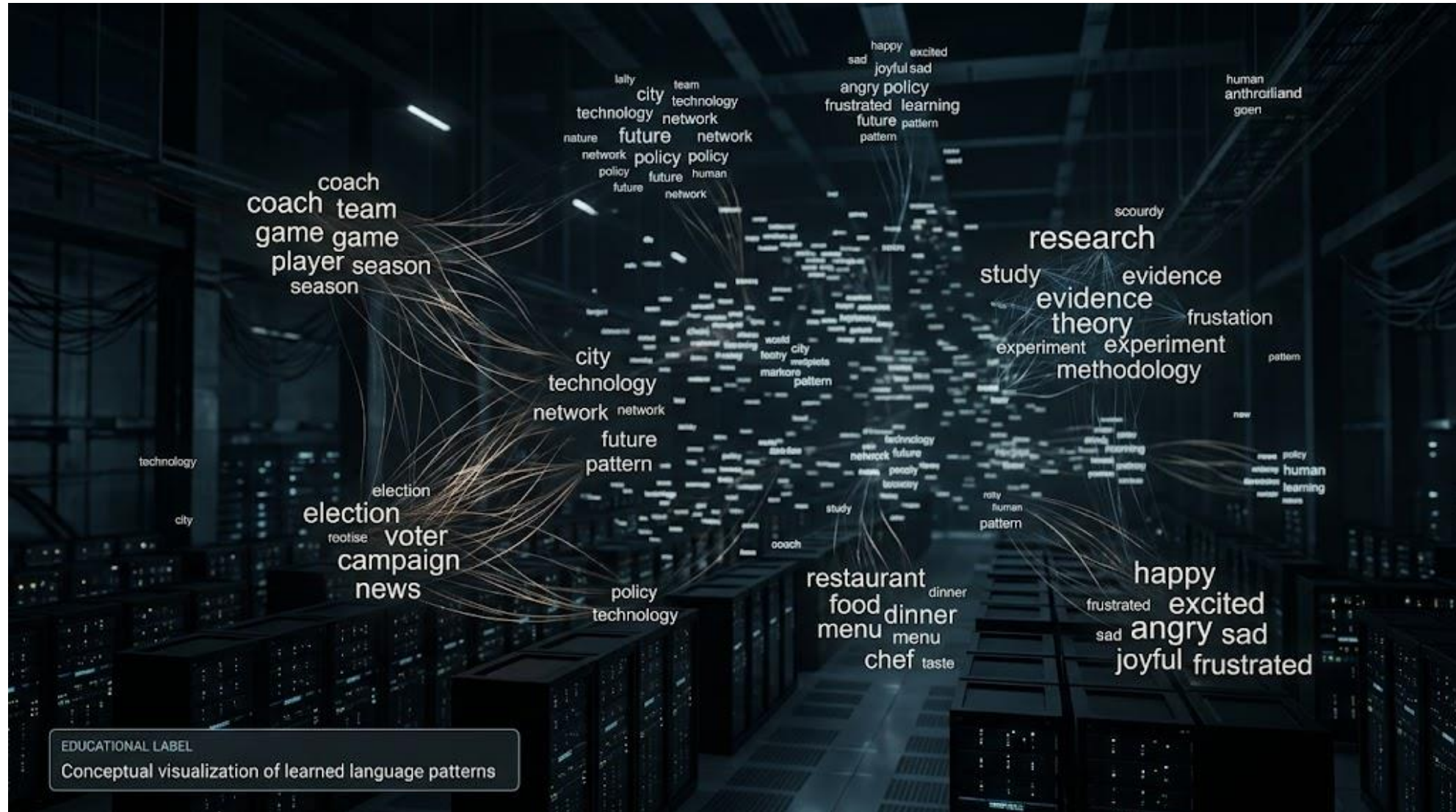
- Text from many sources is broken into smaller pieces called **tokens**.
- The model learns which tokens tend to appear near each other.
- It also learns broader patterns: topics, writing styles, arguments, examples, and associations.

How a Large Language Model (LLM) Learns During Training



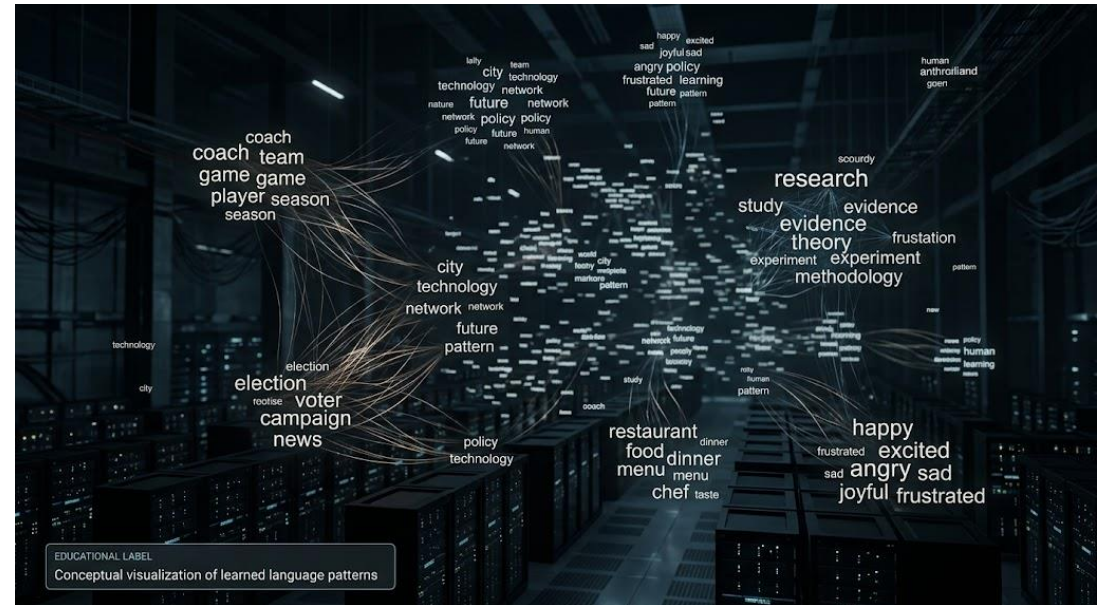
Many Text Sources → Tokens → Learned Patterns & Relationships → Trained LLM

Inside a Trained LLM: Learned Patterns in 3D



2. User Time: From Prompt to Response

- You enter a prompt (AKA input).
- The prompt is also broken into tokens.
- The model processes those tokens using the **patterns and relationships it learned during training**.
- It generates a response **one token at a time**.



User Time: From Prompt to Response

STAGE 1 — USER PROMPT

“How does social media use influence sports fans’ trust in teams?”



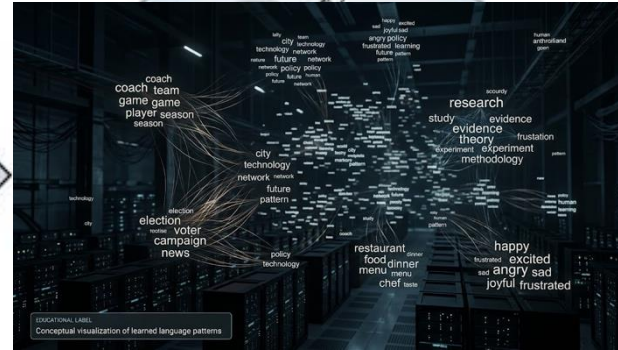
STAGE 2 — TOKENS

“How” “does”
“social media”
“influence” “sports”
“fans” “trust”
“trust” “teams”

“The prompt becomes tokens.”

STAGE 3 — LEARNED PATTERNS

TRAINED LANGUAGE MODEL



TRAINED
LANGUAGE
MODEL

PATTERN
ACTIVATION

“The model processes your prompt using patterns learned during training.”

STAGE 4 — GENERATED RESPONSE

“One possible research question is...”

“One possible research question is... How does social media engagement...”

“One possible research question is... How does social media engagement... relate to sports fans’ trust in professional teams?”

“The model generates a unique response piece by piece.”

PROMPT → TOKENS → LEARNED PATTERNS → GENERATED RESPONSE

So...What's a Transformer?

- Transformer = the technology underneath both training and user time
- The **Transformer** is the backbone behind many modern LLMs.
- ChatGPT, Claude, and Gemini all use Transformer-based models.

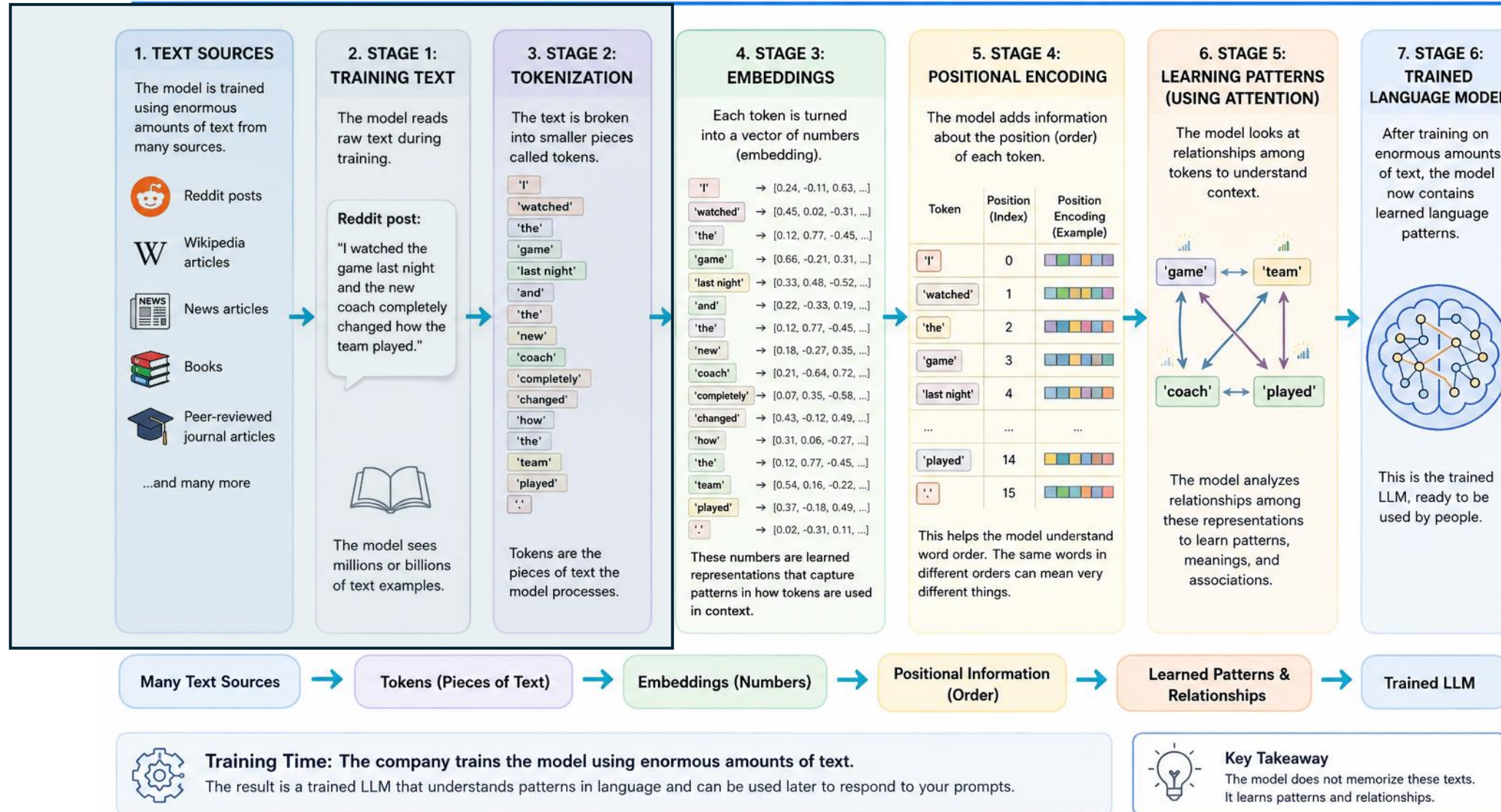
What Happens Inside the Transformer During Training?

Let's Look a Little Deeper at How It Works

Step 1: Break A Sentence Into Pieces (“Tokens”)

How a Large Language Model (LLM) Learns During Training

From raw text to a trained model that understands patterns in language

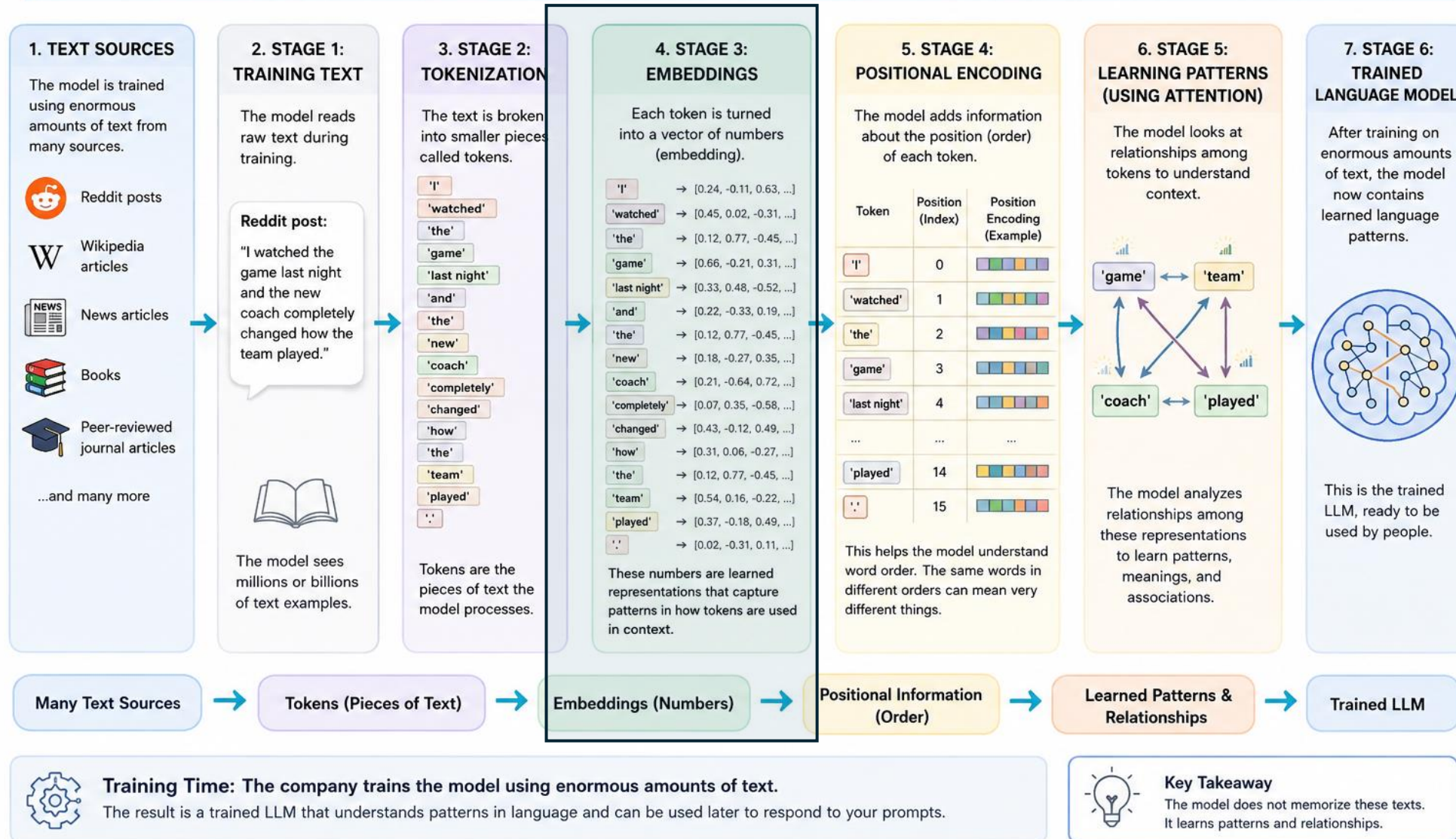


Token = a piece of text the model works with

STEP 2: Turn Pieces Into Numbers (“Embeddings”)

How a Large Language Model (LLM) Learns During Training

From raw text to a trained model that understands patterns in language



Embedding = a numerical representation of a token.

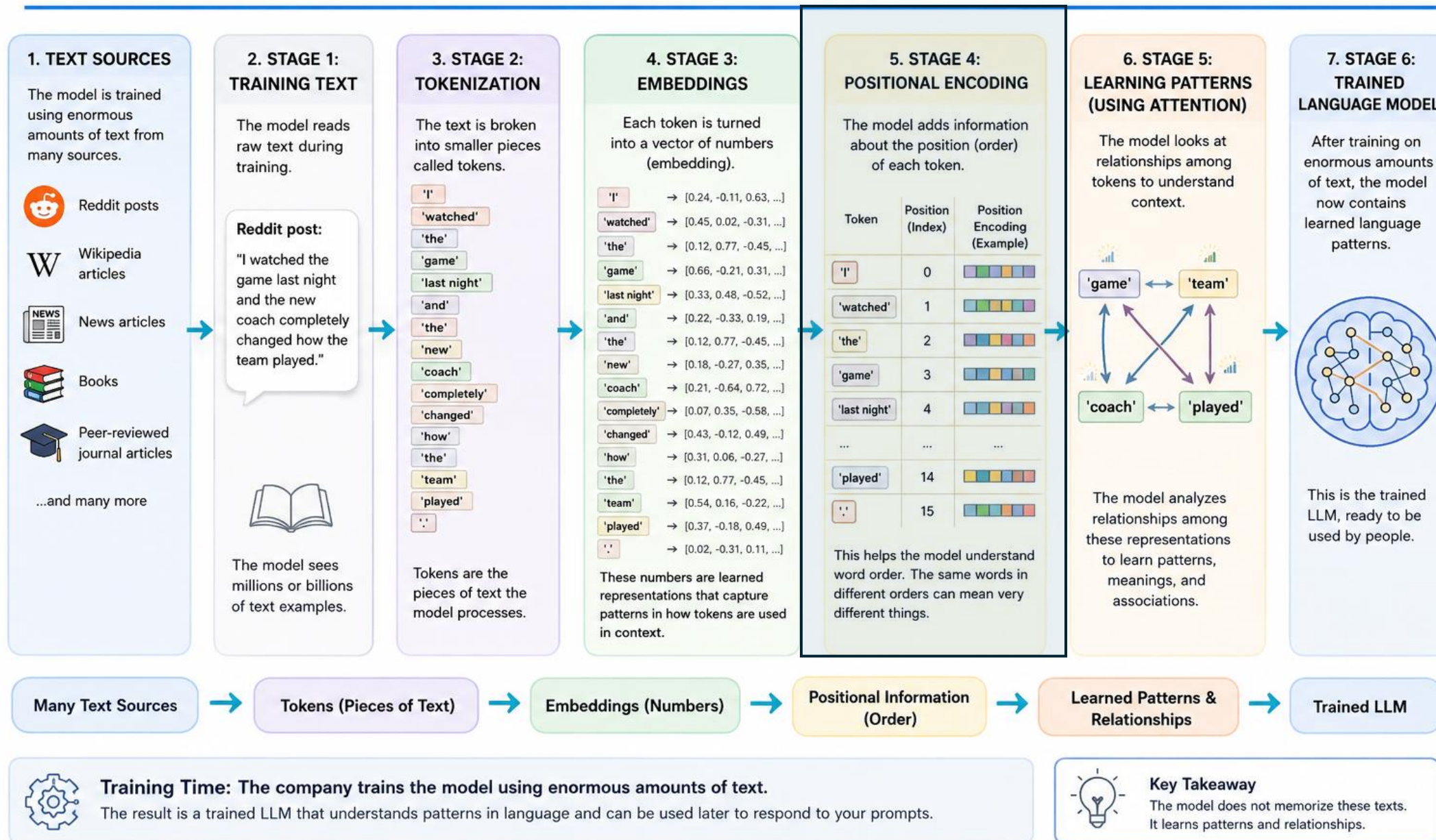
- During training, these representations are learned and adjusted as the model encounters enormous amounts of language.

STEP 3: Order Matters (“Positional Encoding”)

- “**I watched the game last night...**” vs. “**The game watched me last night**”
- Same words. **Different meaning.**
- The model needs information about where tokens appear.

How a Large Language Model (LLM) Learns During Training

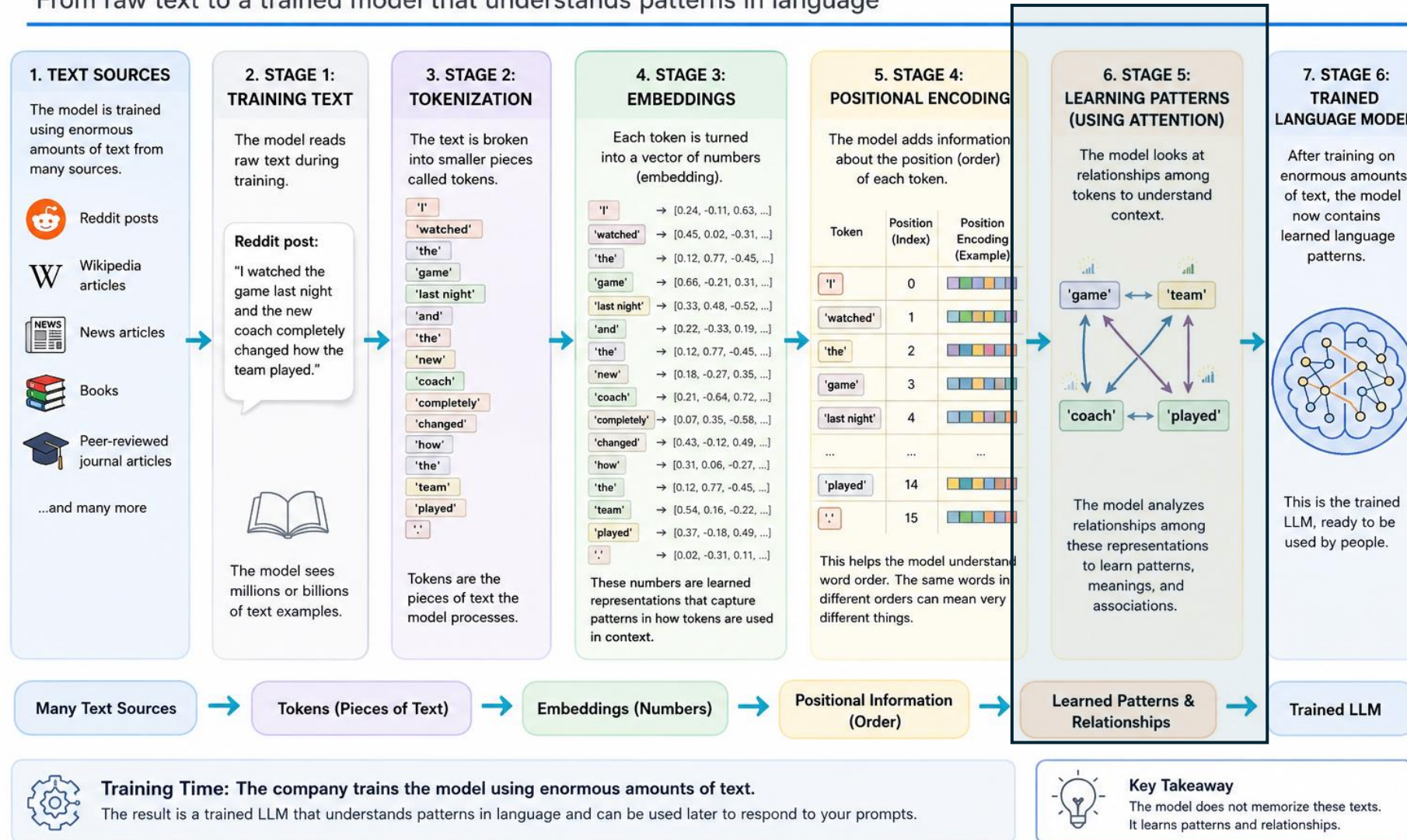
From raw text to a trained model that understands patterns in language



STEP 4: Words Relate to Other Words (“Attention”)

How a Large Language Model (LLM) Learns During Training

From raw text to a trained model that understands patterns in language



Attention helps the model identify which parts of the text are important to each other.

- **team ↔ game**
- **coach ↔ played**
- **game ↔ coach**



What Comes Out of Training?

After training, the company now has A TRAINED LLM.

It has learned enormous numbers of patterns and relationships from its training data.

Training is done. Now we can use the model.

Now: User Time

- This is what happens when YOU use ChatGPT.
- **You type:** “Help me develop a research question about sports fans and social media.”
- The trained model processes your prompt.

Wait...The Prompt Becomes Tokens Again?

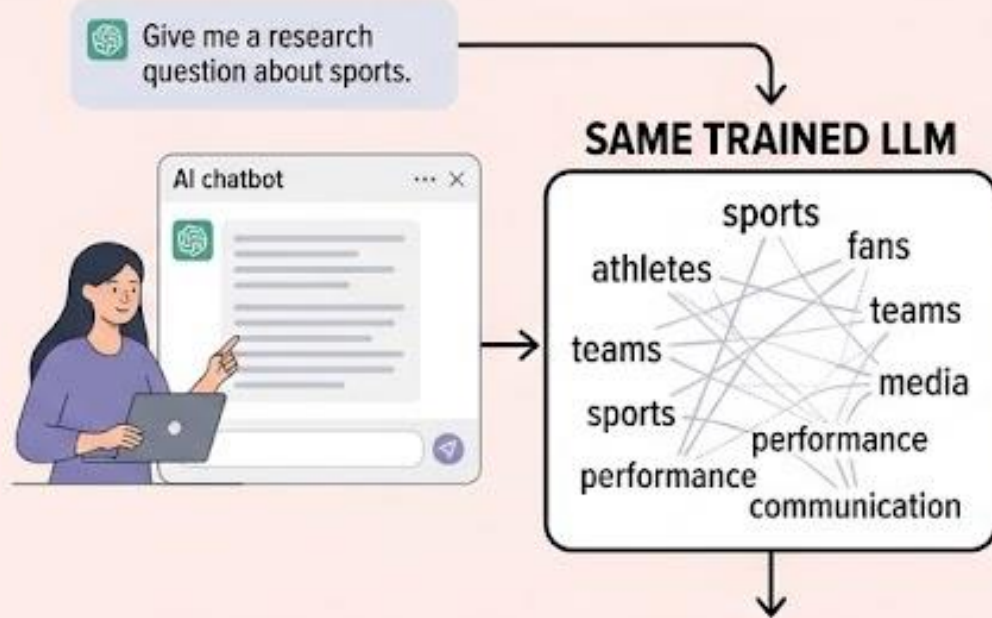
- **Yes!** Your prompt: “Help me develop a research question about sports fans.”
- → Help | me | develop | a | research | question | sports | fans
- **The same basic ideas are used again.**
 - **Tokens** → pieces
 - **Embeddings** → numerical representations
 - **Position** → order
 - **Attention** → relationships/context

BUT:

- The model is **NOT** learning from scratch now.
- It is **USING** what it already learned.

USER TIME: Why Does Your Prompt Matter?

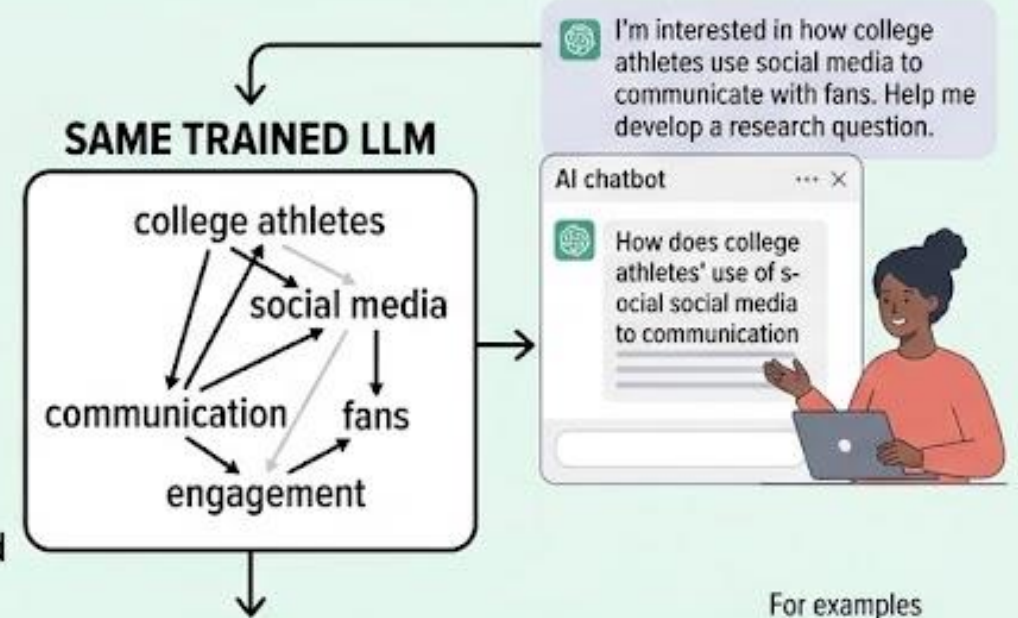
PROMPT A: Very Little Context



"How does social media use affect sports fans' engagement with professional teams?"

Broad prompt → Broad context → One possible response

PROMPT B: More Context



For examples

"How does college athletes' use of social media shape their communication and engagement with fans?"

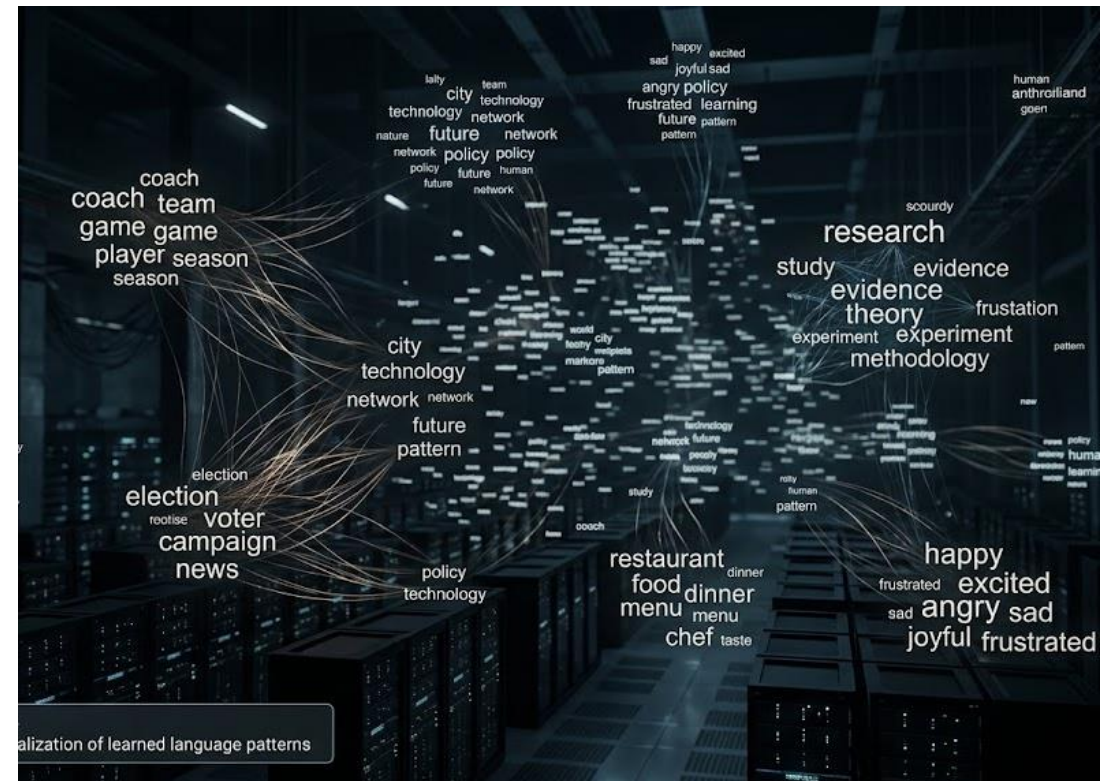
More specific prompt → More specific context → Different possible response

DIFFERENT WORDS → DIFFERENT CONTEXT → POTENTIALLY DIFFERENT RESPONSE

This is one reason prompting matters.

Response generated one token at a time.

- The model predicts **what token is likely to come next**
- The prediction is based on **patterns learned during training**
- It adds a token and **repeats the process**
- **Example:**
- “The coach talked to the ____”
 - **team** ← more likely
 - **players** ← likely
 - **banana** ← very unlikely

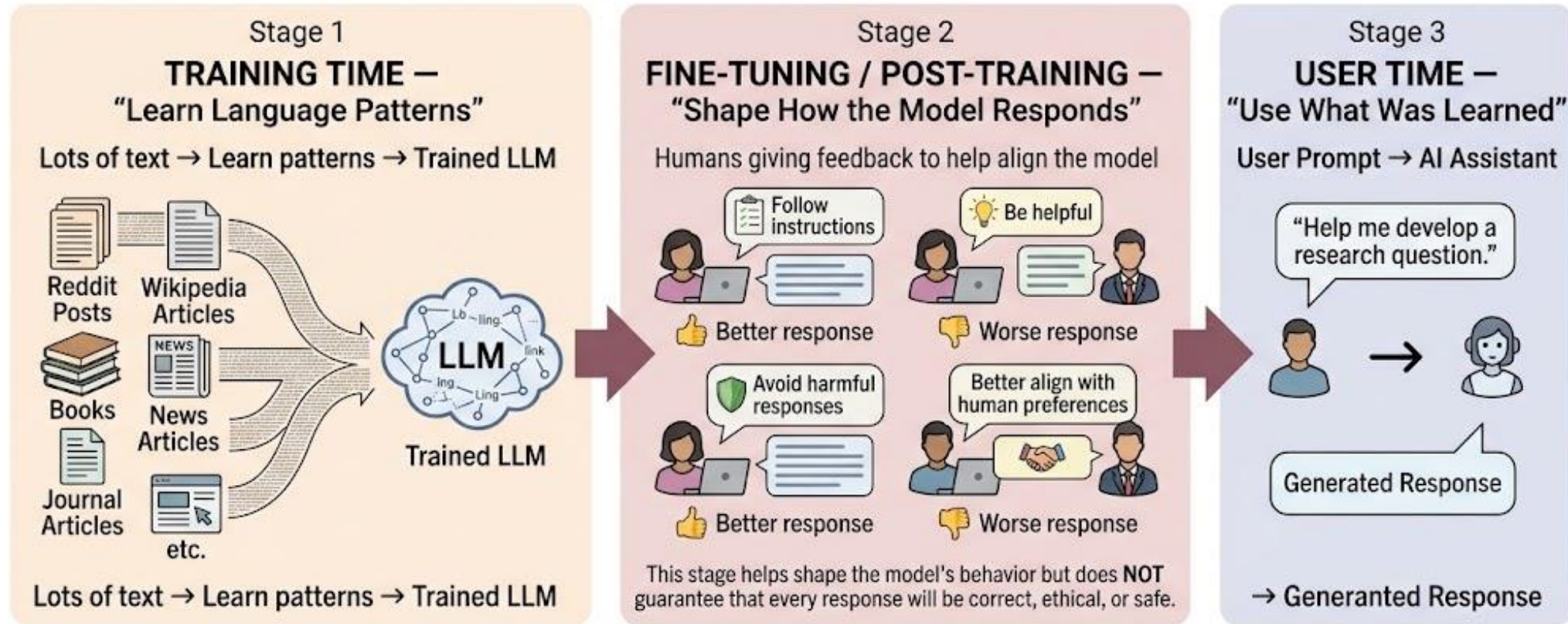


Group Activity: Explain What Happens Behind the Screen

- **Imagine a researcher types this into ChatGPT: “Help me develop a research question about how social media affects sports fans.”**
- **1. Is this TRAINING TIME or USER TIME? Why?**
- **2. What happens to the prompt first?**
Hint: Think about what happens to the words you typed.
- **3. What happens to those pieces of text next?**
Hint: Can a computer work directly with words like we do?
- **4. How does the model know the order of the words?**
Why might order matter?
- **5. How does the model use the other words in the prompt to understand context?**
Hint: Think about attention.
- **6. How does the model begin creating its response?**
- **7. Where did the model learn the language patterns it is using?**
Is it learning them now?

In the real world, there is one more step from a trained LLM to a chatbot

From Training an LLM to Using an AI Assistant



TRAINING → FINE-TUNING / POST-TRAINING → USER TIME
Learn language → Shape behavior → Respond to users

To Sum Up

- **AI** is the broadest category — technologies that can perform tasks that seem “smart,” such as self-driving systems, recommendation systems, Face ID, or voice assistants.
- **Generative AI** is AI that can create new content, such as text, images, audio, video, or code.
- **Large Language Models (LLMs)** are a type of generative AI focused on language.

To Sum Up

- Examples of LLM-based systems include **ChatGPT, Claude, and Gemini**.
- **Transformer** is the main architecture behind many modern LLMs.
- We can think about LLMs in two moments:
 Training time = the model learns patterns
 User time = the trained model uses those patterns to respond



NEXT: Python as a Research Tool

Setting Up Google Colab

